

HORN: Directional Outlier Propagation in Hybrid Reasoners

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Paper under double-blind review

Abstract

This is a living paper shell for a preregistered Mac-first experiment. HORN tests whether attention-to-SSM and SSM-to-attention boundaries differ in outlier magnitude and quantization-noise sensitivity. No positive result or native systems claim is made yet.

1 Current Gate

The branch must pass activation-magnitude, quantization-noise propagation, and cross-model/control gates before GPU validation. H1 first tests directional activation statistics, H2 tests matched FP4-equivalent noise propagation, and H3 requires the direction to survive hybrid-model repeats while weakening on pure-attention and pure-Mamba controls.

2 Mac Artifact Status

A deterministic synthetic H1 packet validates the directional-boundary readout. It reports SSM-to-attention over attention-to-SSM max ratio 3.775 and kurtosis ratio 7.139, producing `SYNTHETIC_PASS_REAL_BOUNDARY_DUMPS_NEXT`. This is synthetic-only: it is not model evidence, not a quantization-noise result, and not GPU evidence. The metadata-only model eligibility packet identifies live hybrid targets and records architecture-map hashes for admissible packets, but no target weights are cached repo-locally, so HORN has not yet produced real boundary rows on the current Mac.

3 Next Evidence

The next admissible row is a real boundary activation dump from a hybrid model. HORN remains a control branch unless real H1/H2/H3 gates show consistent directional asymmetry that pure-attention and pure-Mamba controls do not share. Pure-architecture controls are an H3 requirement, not a blocker for the first H1 boundary-activation packet.

4 Admissible Real Packet

The real H1 packet must use the shared config-derived architecture maps rather than substring-only module classification. Each row must report `model_id`, `prompt_id`, adjacent layer IDs, `direction`, `boundary_index`, normalization placement, `max_abs`, `rms`, `kurtosis`, and `control_type`. The required controls are boundary rows, non-boundary adjacent rows, and permuted-direction rows. A valid packet must cover both `attention->ssm` and `ssm->attention`; each permuted-direction row must match an observed boundary tuple with the same prompt ID and normalization positions while flipping its direction. It must also include at least 12 fixed prompt IDs unless the packet records an explicit resource limit. Any resource-limited packet must use a `RESOURCE_LIMITED_NOT_PROMOTABLE` decision. The packet must include `prompt_ids.hash`, `architecture_map.hash`, and a summary with prompt count, boundary directions, selected H1 ratio, selected H1 CI lower bound, support fraction, non-boundary control ratio, and permuted-direction ratio, then pass `experimental.shared.check_gate_packet --mode`

real --project horn. The checker recomputes these H1 fields from raw rows and rejects stale or hand-filled summaries. Saved tensor packets should be converted with `experimental.shared.hybrid_trace_packet_builder` before validation.

Required control	Reviewer objection addressed
Explicit boundary map	Avoids substring artifacts.
Non-boundary adjacent pair	Separates boundary effects from depth effects.
Permuted direction label	Tests whether direction carries signal.
Matched normalization side	Controls RMSNorm/LayerNorm placement.
Pure-architecture control	Prevents generic outlier claims.

Table 1: Controls required before HORN can claim directional hybrid asymmetry.

5 Reviewer Packet

The reviewer-facing claim boundary is tracked in `experimental/horn/paper/reviewer_pack.md`. It records that HORN is not camera-ready as a method or measurement paper until real H1–H3 evidence exists.